

Microeconomics: Choice, Preference, and Utility

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- Microeconomics provides a framework for studying how **institutions** (including markets) shape the utilization of scarce **resources** through the behavior they induce among economic **agents**.

Economic environments

- At an abstract level, an economic environment has three key ingredients:
 - ① a set of agents,
 - ② a set of possible outcomes,
 - ③ agents' preferences over these outcomes.
- An *institution* specifies the rules governing how agents may act and interact.
- Agents make decisions within these rules, and their decisions jointly determine the resulting outcome.
- We study how different institutions shape behavior and evaluate the outcomes they produce.

This lecture

- We introduce three formal objects used to model economic agents:
 - ① choice functions,
 - ② preference relations,
 - ③ utility functions.
- We will also study when these different ways of modeling behavior are consistent with one another.
- Reference: Chapters 1 and 2 of Osborne and Rubinstein.

Set of possibilities: Alternatives

- Let X denote a fixed set of *alternatives*.
- X may represent possible societal outcomes, such as:
 - ① candidates in an election,
 - ② possible locations for building a library,
 - ③ action profiles when two players play rock-paper-scissors,
 - ④ assignments of students to dorm rooms.
- Alternatively, X may represent outcomes for an individual, such as:
 - ① seats in a cinema hall,
 - ② dorm rooms,
 - ③ lotteries in a casino,
 - ④ bundles that can be purchased with INR 1,000.
- We want to formally capture how an agent evaluates these alternatives.

1. Choice functions

- Given X , let $P(X)$ denote the set of all non-empty subsets of X .
- A *choice function* on X is a function

$$c : P(X) \rightarrow P(X)$$

such that

$$c(A) \subseteq A \quad \text{for every } A \in P(X).$$

- If an agent is offered the set of alternatives A , then $c(A)$ is the set of acceptable choices from A : the agent is willing to choose any one alternative in $c(A)$.
- Thus, a choice function directly describes the agent's observable choice behavior.

Illustration

- Suppose

$$X = \{a, b, c\}.$$

- Then

$$P(X) = \{\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}.$$

- How many different choice functions are there on X ?
- Choice functions allow us to model very general forms of behavior.
- But some possible choice functions may display behavior that we would regard as inconsistent.

Example

- Suppose

$$X = \{\text{tea}, \text{coffee}, \text{juice}\}.$$

- An agent chooses

- 1 $\{\text{coffee}\}$ from $\{\text{tea}, \text{coffee}\}$,
- 2 $\{\text{juice}\}$ from $\{\text{coffee}, \text{juice}\}$,
- 3 $\{\text{tea}\}$ from $\{\text{juice}, \text{tea}\}$.

- The choices form a cycle:

$$\text{coffee} \succ \text{tea}, \quad \text{juice} \succ \text{coffee}, \quad \text{tea} \succ \text{juice}.$$

- Such behavior may, for example, leave an agent vulnerable to a money-pump argument.

Sen's α : contraction consistency

- Suppose

$$c(\{a, b, c\}) = \{a\}, \quad c(\{a, b\}) = \{b\}.$$

- Alternative a is chosen when $\{a, b, c\}$ is available, but stops being chosen when the unavailable alternative c is removed.
- One might require that an alternative that is chosen from a larger menu remain chosen when the menu shrinks, as long as that alternative remains available.
- A choice function $c : P(X) \rightarrow P(X)$ satisfies *Sen's α* if

$$a \in A \subseteq B \quad \text{and} \quad a \in c(B) \quad \implies \quad a \in c(A).$$

Sen's β : expansion consistency

- Suppose

$$c(\{a, b\}) = \{a, b\}, \quad c(\{a, b, c\}) = \{a\}.$$

- Alternatives a and b are both chosen from the smaller menu.
- If the menu expands and a remains chosen, b should also be chosen.
- A choice function $c : P(X) \rightarrow P(X)$ satisfies *Sen's β* if

$$a, b \in A \subseteq B, \quad a, b \in c(A), \quad a \in c(B) \implies b \in c(B).$$

- Suppose

$$c(\{a, b, c\}) = \{a\}, \quad c(\{a, b, d\}) = \{b\}.$$

- When a is chosen from the first menu, b is available but not chosen.
- If b is chosen from another menu that also contains a , then a must also be chosen.
- A choice function $c : P(X) \rightarrow P(X)$ satisfies the *Weak Axiom of Revealed Preference (WARP)* if

$$a, b \in A \cap B, \quad a \in c(A), \quad b \in c(B) \implies a \in c(B).$$

2. Preferences

- A preference describes how an agent compares different alternatives.
- We require that the agent be able to compare every pair $x, y \in X$. For any pair (x, y) , the agent must regard one of the following as true:
 - 1 x is preferred to y ,
 - 2 y is preferred to x ,
 - 3 x and y are equally desirable.
- We also require consistency across comparisons. For example, if x is preferred to y and y is preferred to z , then x should be preferred to z .
- We formalize preferences using binary relations.

Preference relation

- A *preference relation* on X is a binary relation

$$\succsim \subseteq X \times X$$

that is:

- ① *complete*: for every $x, y \in X$,

$$x \succsim y \quad \text{or} \quad y \succsim x \quad \text{or both;}$$

- ② *transitive*: for every $x, y, z \in X$,

$$x \succsim y \quad \text{and} \quad y \succsim z \quad \implies \quad x \succsim z.$$

- The statement

$$x \succsim y$$

means that the agent regards x as at least as desirable as y .

Strict preference and indifference

- Given a preference relation $\succsim$ on X :

- we say that x is *strictly preferred* to y , written $x \succ y$, if

$$x \succsim y \quad \text{and} \quad y \not\succsim x.$$

- we say that x and y are *indifferent*, written $x \sim y$, if

$$x \succsim y \quad \text{and} \quad y \succsim x.$$

- Some useful properties:

- $\succ \cup \sim = \succsim$
- $\succ \cap \sim = \emptyset$
- $\succ$ is transitive.

From preferences to choices

- Suppose an agent has a preference relation $\succsim$ on X .
- If the agent is offered a menu A , a natural prediction is that the agent chooses the alternatives that are at least as desirable as every other alternative in A .
- Thus, preferences generate the choice rule

$$c(A) = \{x \in A : x \succsim y \text{ for every } y \in A\}.$$

- Can observed choice behavior always be explained in this way?

- A choice function $c : P(X) \rightarrow P(X)$ is *rationalizable* if there exists a preference relation $\succsim$ on X such that, for every $A \in P(X)$,

$$c(A) = \{x \in A : x \succsim y \text{ for all } y \in A\}.$$

- In other words, the observed choices can be explained as maximizing some complete and transitive preference relation.
- If a choice function is rationalizable, must it satisfy our consistency axioms?
- Conversely, if a choice function satisfies the consistency axioms, must it be rationalizable?

Theorem 1.

Let X be a finite set and let

$$c : P(X) \rightarrow P(X)$$

be a choice function. The following statements are equivalent:

- ① c satisfies Sen's α and Sen's β ;
- ② c satisfies WARP;
- ③ c is rationalizable.

Sen's axioms imply WARP

Suppose c satisfies Sen's α and Sen's β .

- ① Let $A, B \in P(X)$ and suppose

$$a, b \in A \cap B, \quad a \in c(A), \quad b \in c(B).$$

- ② By Sen's α ,

$$a, b \in c(A \cap B).$$

- ③ Apply Sen's β to

$$A \cap B \subseteq B.$$

Since $a, b \in c(A \cap B)$ and $b \in c(B)$, it follows that

$$a \in c(B).$$

Hence c satisfies WARP.

WARP implies rationalizability

Suppose c satisfies WARP.

- 1 Define a binary relation $\succsim$ on X by

$$x \succsim y \iff x \in c(\{x, y\}).$$

- 2 Show that $\succsim$ is a preference relation:

- 1 $\succsim$ is complete;
- 2 $\succsim$ is transitive.

- 3 Show that, for every $A \in P(X)$,

$$c(A) = \{x \in A : x \succsim y \text{ for every } y \in A\}.$$

- 4 Therefore, c is rationalizable.

Rationalizable implies Sen's axioms

Suppose c is rationalized by $\succsim$:

$$c(A) = \{x \in A : x \succsim y \text{ for all } y \in A\}.$$

① **Sen's α .** Suppose

$$a \in A \subseteq B \quad \text{and} \quad a \in c(B).$$

Then

$$\begin{aligned} a \in c(B) &\implies a \succsim x \text{ for all } x \in B \\ &\implies a \succsim x \text{ for all } x \in A \\ &\implies a \in c(A). \end{aligned}$$

Rationalizable implies Sen's axioms

2 **Sen's β .** Suppose

$$a, b \in A \subseteq B, \quad a, b \in c(A), \quad a \in c(B).$$

Since $b \in c(A)$ and $a \in A$,

$$b \succcurlyeq a.$$

Moreover,

$$\begin{aligned} a \in c(B) &\implies a \succcurlyeq x \text{ for all } x \in B \\ &\implies b \succcurlyeq x \text{ for all } x \in B && \text{(by transitivity)} \\ &\implies b \in c(B). \end{aligned}$$

Hence c satisfies Sen's α and Sen's β .

3. Utility

- In economics, we often represent preferences using utility functions.
- A function

$$U : X \rightarrow \mathbb{R}$$

represents a preference relation $\succsim$ if

$$x \succsim y \iff U(x) \geq U(y).$$

- Utility maximization is an “as if” model: agents need not literally maximize a utility function when making decisions.
- Rather, observed choice behavior is modeled as if it were generated by maximizing some underlying utility function.
- When can a preference relation be represented by a utility function?

Theorem 2.

Every preference relation on a finite set X can be represented by a utility function

$$U : X \rightarrow \mathbb{R}.$$

- Intuitively, rank the indifference classes from worst to best and assign increasingly larger numbers to them.

Infinite sets

- On an infinite set, a preference relation need not admit a utility representation.
- Some do. For example, on $X = \mathbb{R}$,

$$x \succcurlyeq y \iff x \geq y$$

is represented by

$$U(x) = x.$$

- But consider the lexicographic preference on $X = \mathbb{R}^2$:

$$(x_1, x_2) \succcurlyeq (y_1, y_2)$$

if

$$x_1 > y_1 \quad \text{or} \quad (x_1 = y_1 \text{ and } x_2 \geq y_2).$$

- This preference cannot be represented by a real-valued utility function.
- We therefore need additional regularity conditions.

- The relevant condition is continuity: if $x \succ y$, then sufficiently small changes in x and y do not reverse the ranking.
- Formally, a preference $\succsim$ on X is *continuous* if, whenever $x \succ y$, there exist $\epsilon_1, \epsilon_2 > 0$ such that

$$x' \in N_{\epsilon_1}(x), \quad y' \in N_{\epsilon_2}(y) \implies x' \succ y'.$$

Here,

$$N_{\epsilon}(x) = \{x' \in X : d(x, x') < \epsilon\}.$$

- Lexicographic preferences are not continuous.

Theorem 3 (Debreu).

Let X be a convex subset of $\mathbb{R}^n$ and let $\succsim$ be a continuous preference relation on X . Then there exists a continuous function

$$U : X \rightarrow \mathbb{R}$$

that represents $\succsim$.

- We discussed different ways of modeling economic agents.
- Choice functions provide a very general model, but may also allow for potentially inconsistent or pathological behavior.
- Preference relations capture natural consistency requirements.
- Utility functions provide a convenient representation of preferences, making it easier to analyze economic models using tools such as calculus.